Color in data vis
16 min
July 24th, 2023

Emphasize what you want readers to see with color

Lisa Charlotte Muth

AND THIS THIRD
YOU READ
THIS FIRST
THIS SECOND
Degrees warmer or cooler than August average
-20°F -15° -10° -5° 0 +5° +10° +15°

One of the biggest superpowers of colors might be to lead a reader's eye. By emphasizing certain data points and de-emphasizing others, you can decide what you want readers to see first, second, third, and last. And it works with any color scale: Whether you're showing numerical data in a gradient from yellow to blue, or categories in different hues like purple, yellow, and orange, you can always highlight certain categories, ranges, or values that you consider especially important to your audience or story. This article explains how.

Here's what we'll talk about:

01 **Start by thinking about which information is most important**
02 **Make everything else gray**
03 **Don't use gray like other colors**
04 **Use the same hue for less important categories, but desaturate it**
05 **Don't use new hues for (de-)emphasized categories**
06 **Use more saturated colors for more important categories**
07 **Use saturation to emphasize categories a bit more or less**

Start by thinking about which information is most important.

To emphasize or de-emphasize data points, you first need to decide what's important. What should people take away from looking at your visualization? **What do you want them to see and understand about your data?** What question(s) should readers be able to answer?

I recommend **writing down the answers** if you're just starting out.

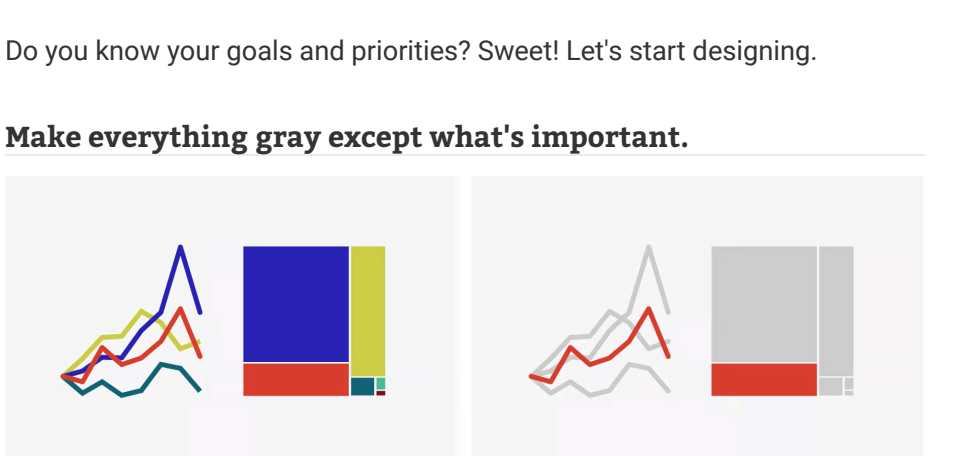
I also recommend prioritizing your goals: What's most important to show — what should people definitely not miss? What point would be great, but not essential, for readers to spot? And what might be your personal darling data point or category, but is probably not of the highest importance to others?

For example, in a chart about life expectancy, your main priority might be to show that *life expectancy improved in all countries over the last hundred years*. It might be less important, but still useful to show that *life expectancy today is higher in countries with a high GDP*. And you might personally really like the fact that *life expectancy has increased faster in countries with a high GDP* — but if you had to decide, you'd say the first two statements are more important for readers to understand.

Prioritizing means making tradeoffs. If everything is emphasized, nothing is — and readers might be so overwhelmed by your visualization that they stop looking at it. **It's better for readers to take away your most important statement than none at all.**

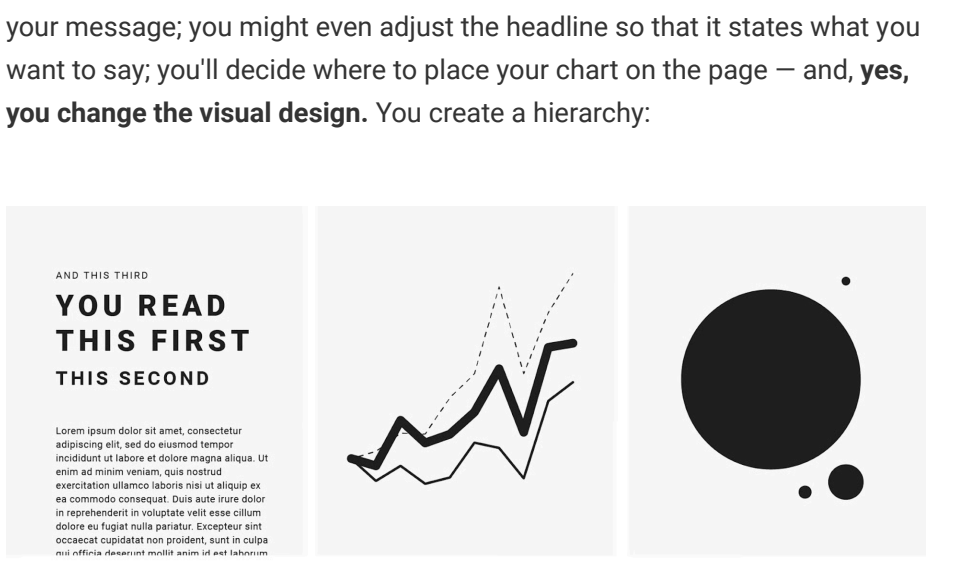
Do you know your goals and priorities? Sweet! Let's start designing.

Make everything gray except what's important.



EVERYTHING IS EMPHASIZED | DEEMPHASIZE WITH GRAY

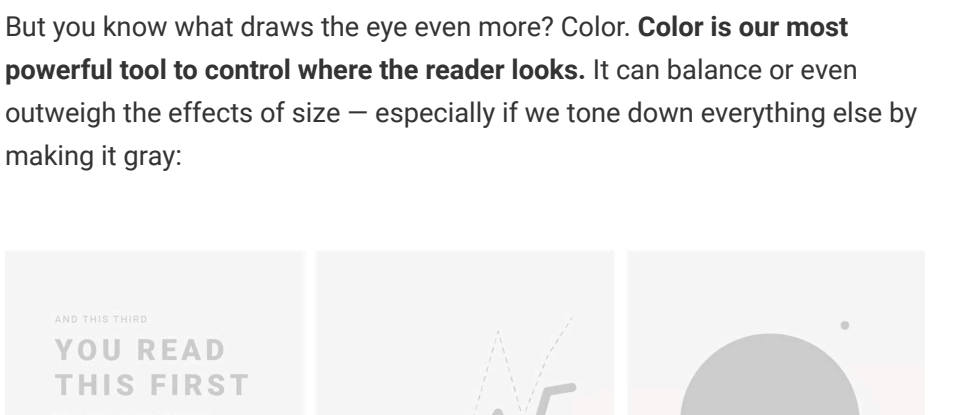
Your priorities will guide **decisions about your visualization.** You'll decide which categories are important to show and which ones can be grouped together or removed; you'll choose a visualization type that brings across your message; you might even adjust the headline so that it states what you want to say; you'll decide where to place your chart on the page — and, **yes, you change the visual design.** You create a hierarchy:



FONT SIZE & STYLE | LINE WIDTH | AREA SIZE

The **bigger** and **bolder** you make something, the more attention your readers will give it. Our eyes are immediately drawn to big things.

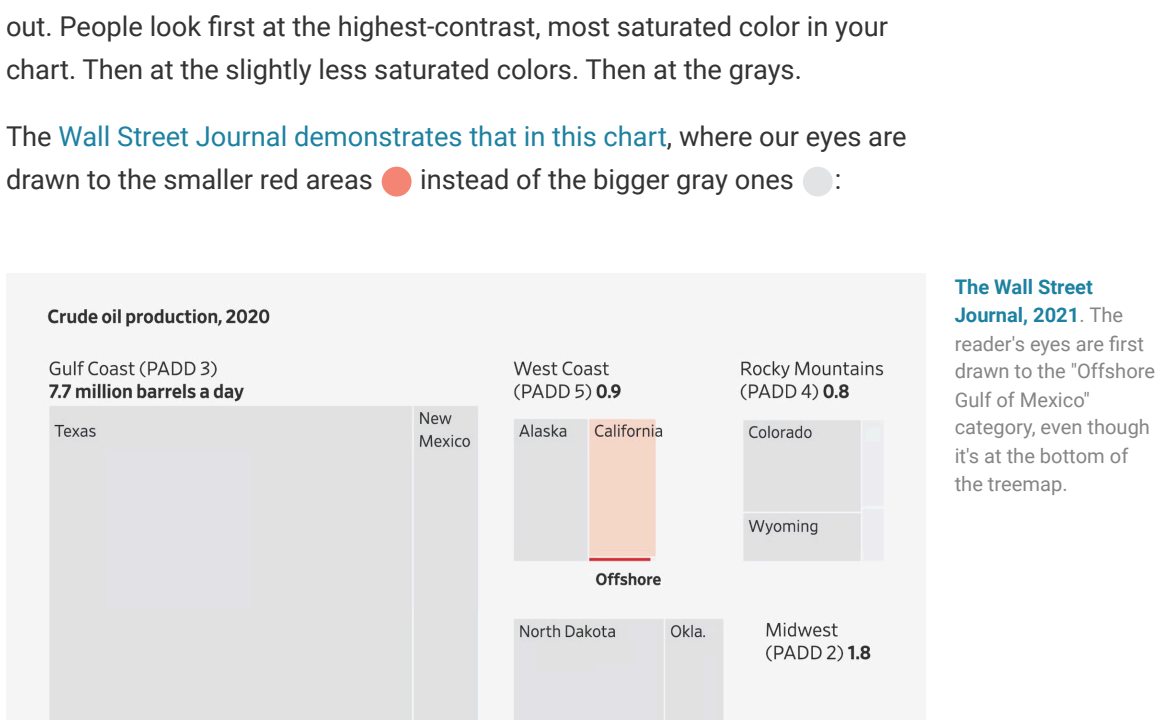
But you know what draws the eye even more? Color. **Color is our most powerful tool to control where the reader looks.** It can balance or even outweigh the effects of size — especially if we tone down everything else by making it gray:



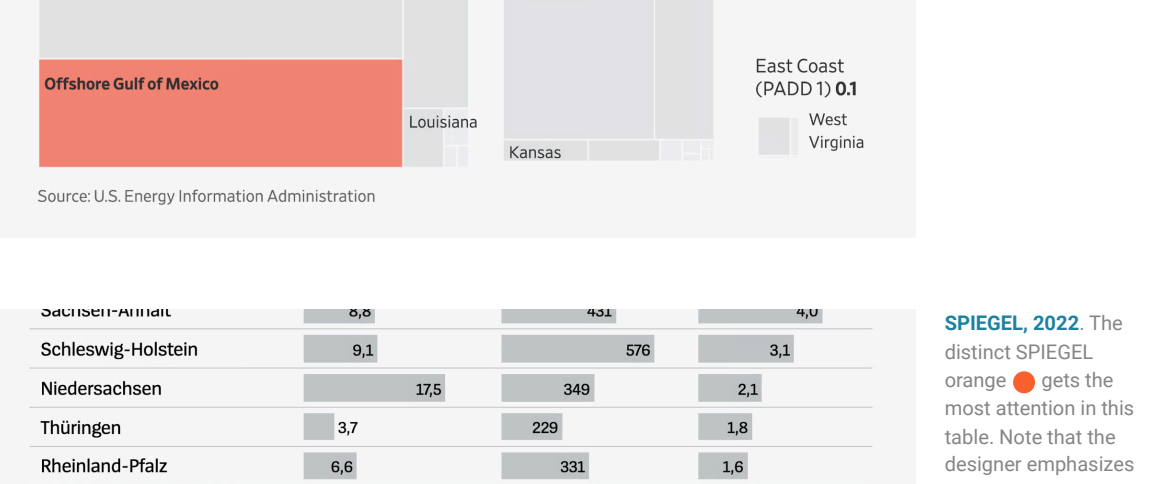
FONT SIZE & STYLE | LINE WIDTH | AREA SIZE

Gray is a storytelling tool: Against gray elements, colored ones will stick out. People look first at the highest-contrast, most saturated color in your chart. Then at the slightly less saturated colors. Then at the grays.

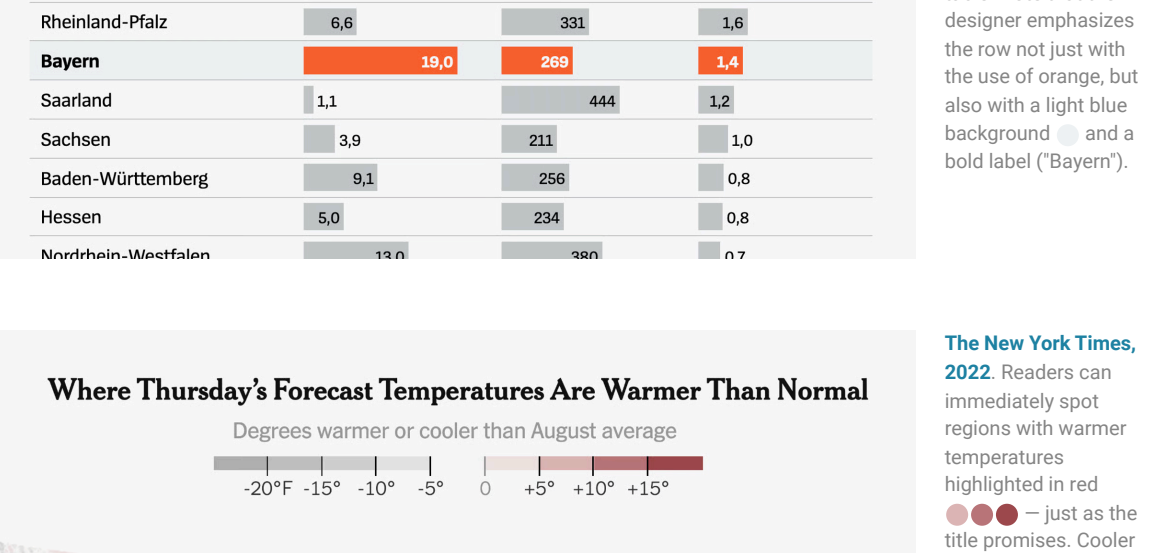
The Wall Street Journal demonstrates that in this chart, where our eyes are drawn to the smaller red areas ● instead of the bigger gray ones ●:



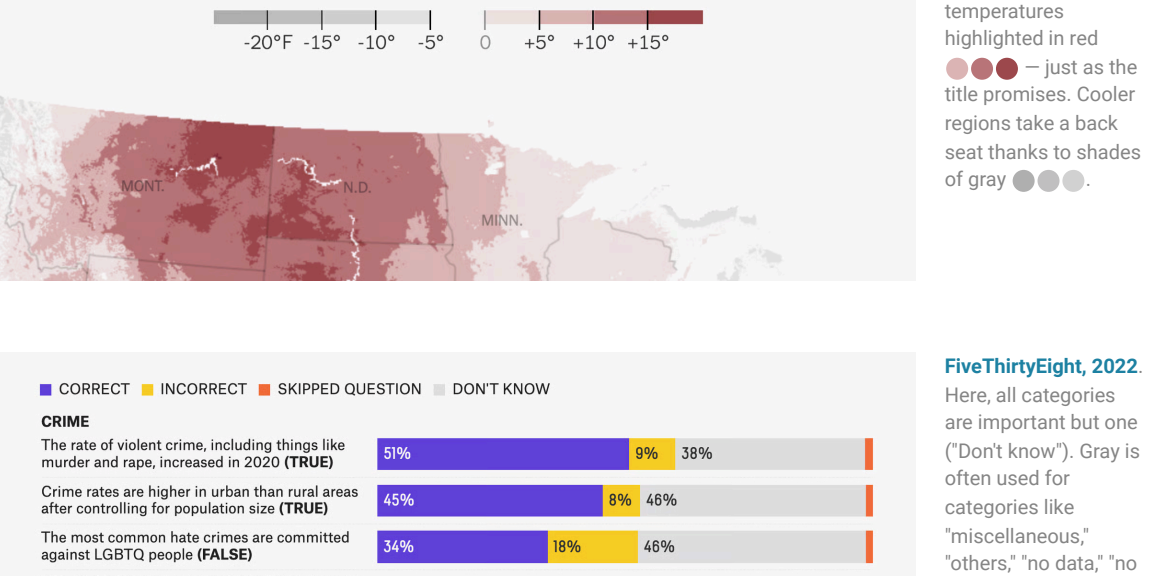
The Wall Street Journal, 2021. The reader's eyes are first drawn to the "Offshore Gulf of Mexico" category, even though it's at the bottom of the treemap.


Schleswig-Holstein
Niedersachsen
Thüringen
Rheinland-Pfalz
Bayern
Saarland
Sachsen
Baden-Württemberg
Hessen
Nordrhein-Westfalen

SPIEGEL, 2022. The distinct SPIEGEL orange ● gets the most attention in this table. Note that the designer emphasizes the row not just with the use of orange, but also with a light blue background ● and a bold label ("Bayern").

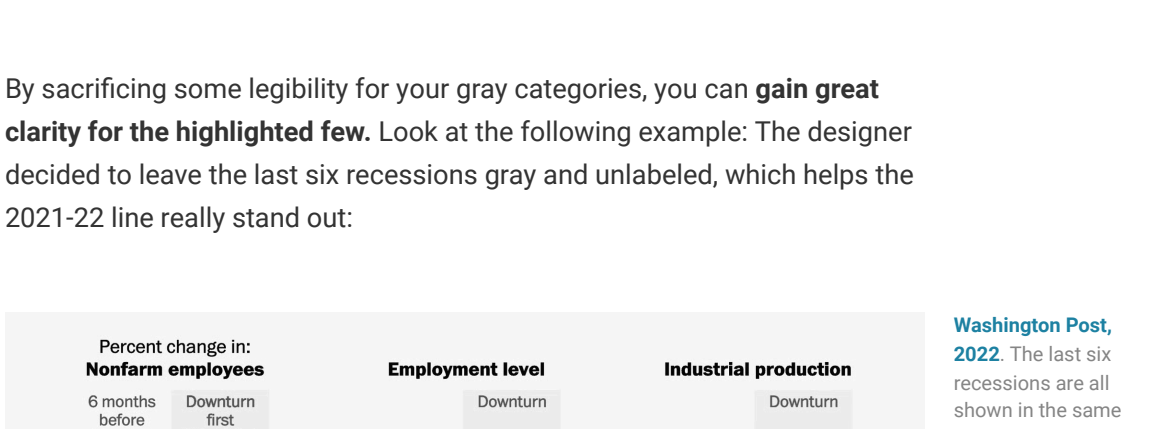

Where Thursday's Forecast Temperatures Are Warmer Than Normal
Degrees warmer or cooler than August average
-20°F -15° -10° -5° 0 +5° +10° +15°

The New York Times, 2022. Readers can immediately spot regions with warmer temperatures highlighted in red ●●● — just as the title promises. Cooler regions take a back seat thanks to shades of gray ●●●.



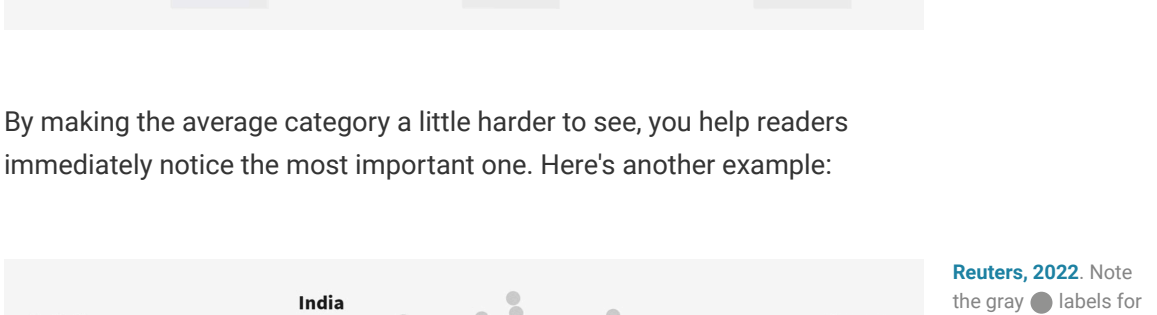
FiveThirtyEight, 2022. Here, all categories are important but one ("Don't know"). Gray is often used for categories like "miscellaneous," "others," "no data," "no answer," or "don't know" — they're almost always the least important.

By sacrificing some legibility for your gray categories, you can **gain great clarity for the highlighted few.** Look at the following example: The designer decided to leave the last six recessions gray and unlabeled, which helps the 2021-22 line really stand out:



Washington Post, 2022. The last six recessions are all shown in the same gray ● and same stroke width, making the current one stand out ●.

By making the average category a little harder to see, you help readers immediately notice the most important one. Here's another example:

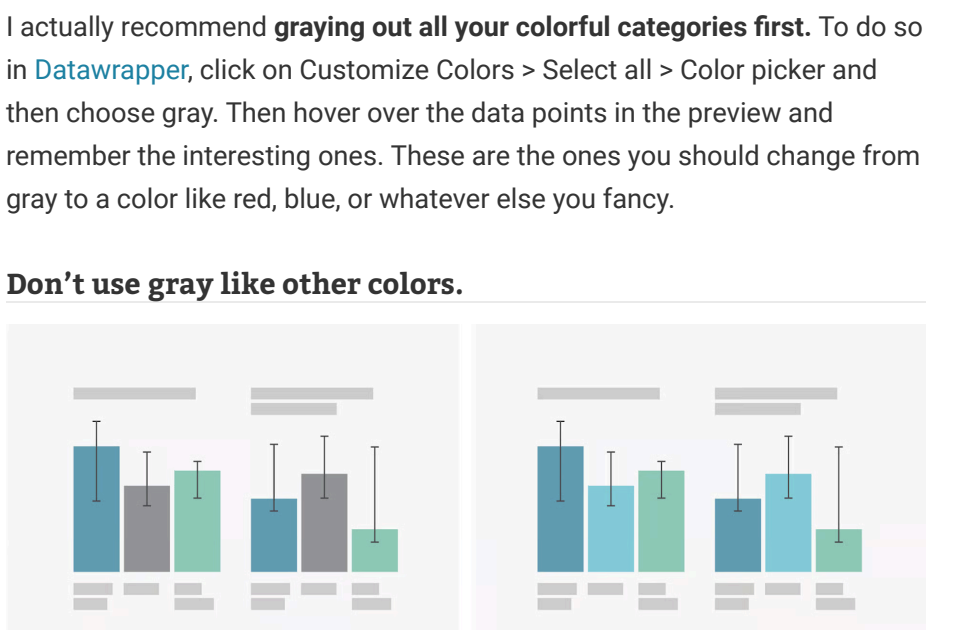


Reuters, 2022. Note the gray ● labels for outliers North Korea, Eritrea, and Norway. They're interesting but less important than India ● and Pakistan ●.

In short: **Goals** ("people should get this message") and **priorities** ("this category is more important than that one") lead to **design decisions** ("I'll make the other categories gray").

I actually recommend **graying out all your colorful categories first.** To do so in Datawrapper, click on Customize Colors > Select all > Color picker and then choose gray. Then hover over the data points in the preview and remember the interesting ones. These are the ones you should change from gray to a color like red, blue, or whatever else you fancy.

Don't use gray like other colors.

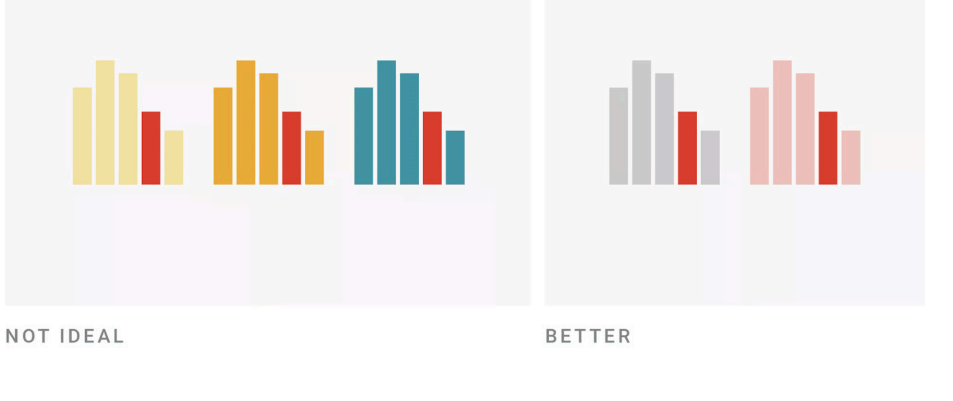


NOT IDEAL | BETTER

As we've seen, gray is special and should be treated as such. By now, many of your readers are used to colorful visualizations that use gray for less important categories. If a category is as important as other, colorful ones, don't make it gray.

You might want to use gray as a normal category color because you "ran out of colors" in a chart with lots of categories. Or maybe the data vis style guide you're following is limited and doesn't offer enough colors for your needs. If that's the case, get in touch with the people who can change that in your organization (for your current project and also long-term).

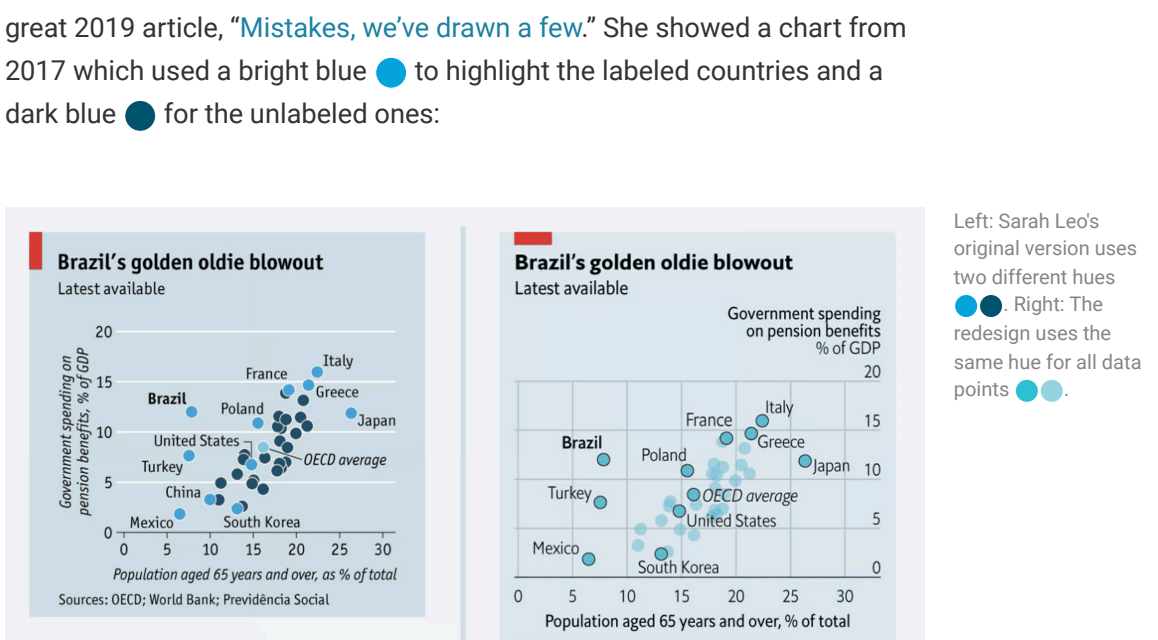
Use the same hue for less important categories, but desaturate it



EVERYTHING IS EMPHASIZED | DEEMPHASIZE WITH A LESS SATURATED HIGHLIGHT COLOR

So far, we've only looked at gray as a color for unimportant data points. But there are other options: For example, you can use **a less saturated version of the highlight color** for all other categories.

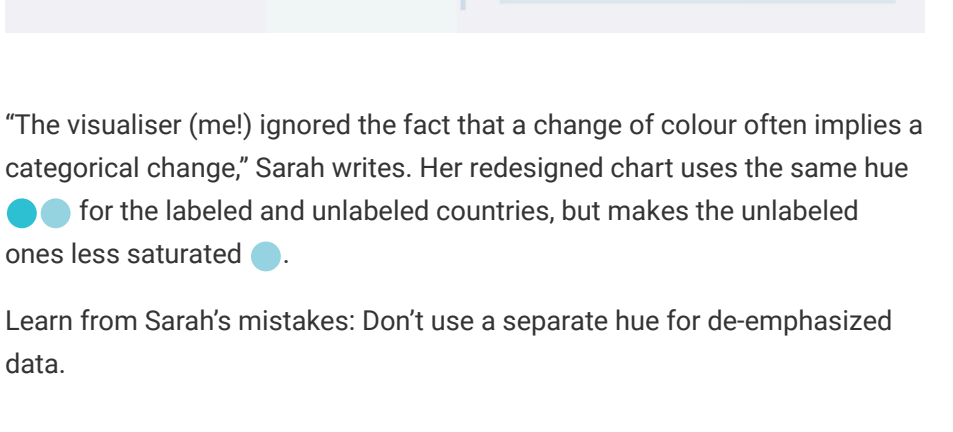
Here, you basically switch gray for a bit more color:



The Economist, 2022. The US average line is colored in a saturated, strong blue ● while the lines for states are toned down to a lighter, less saturated blue ●.

Make sure it really is simply a less saturated version of the highlight color. Don't do this:

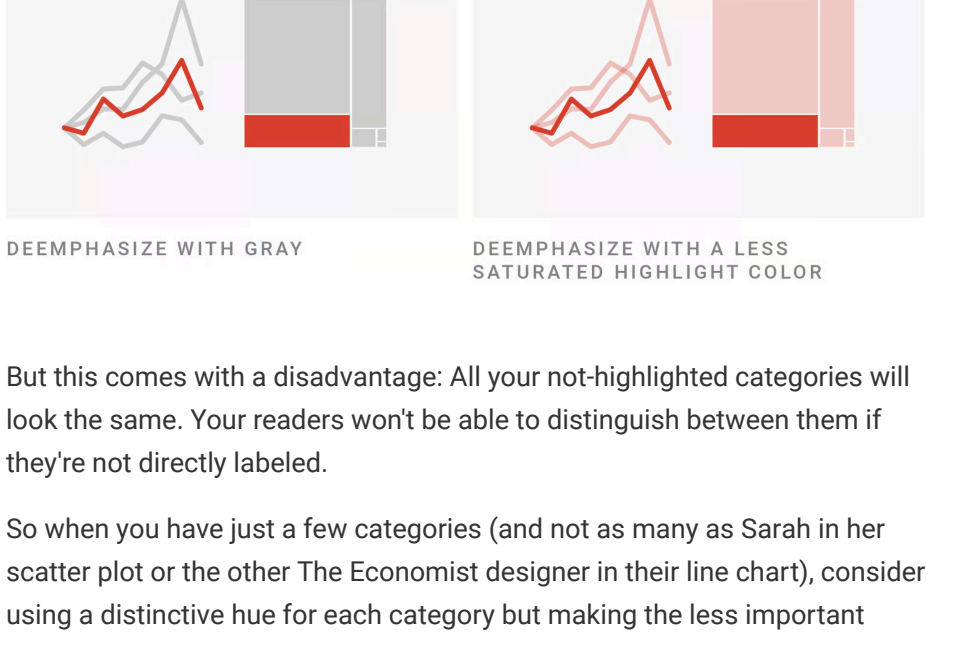
Don't use new hues for deemphasized categories.



NOT IDEAL | BETTER

To be on the safe side, stick to gray or desaturation for de-emphasized categories. Don't get experimental here — another hue will communicate "this is a separate category," not "these are all less important categories,"

Sarah Leo, visual data journalist at The Economist, wrote about this in her great 2019 article, "Mistakes, we've drawn a few." She showed a chart from 2017 which used a bright blue ● to highlight the labeled countries and a dark blue ● for the unlabeled ones:



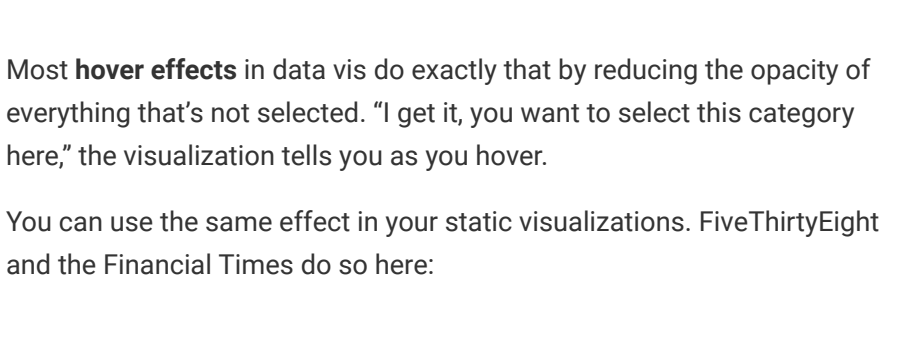
Left: Sarah Leo's original version uses two different hues ●●. Right: The redesign uses the same hue for all data points ●●.

"The visualiser (me!) ignored the fact that a change of colour often implies a categorical change," Sarah writes. Her redesigned chart uses the same hue ●● for the labeled and unlabeled countries, but makes the unlabeled ones less saturated ●.

Learn from Sarah's mistakes: Don't use a separate hue for de-emphasized data.

Use more saturated colors for more important categories

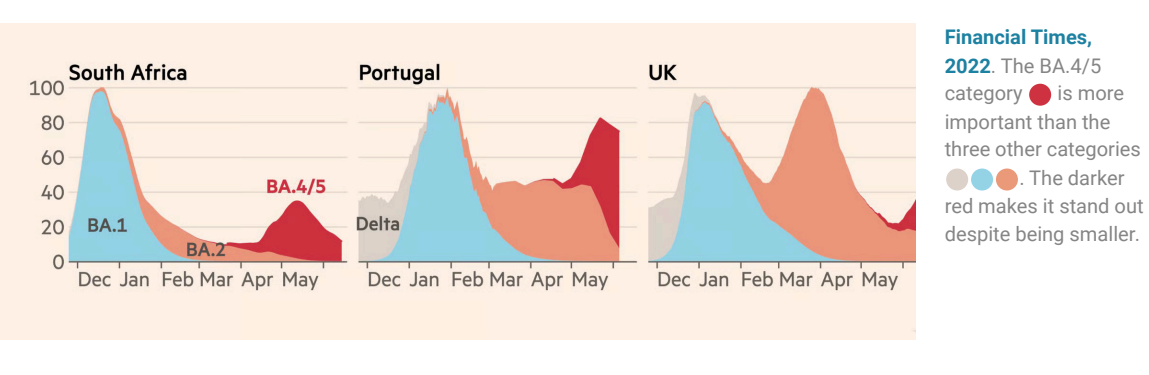
So far, we've looked at ways to highlight one or two categories by giving all the others the same (gray or desaturated) color.



DEEMPHASIZE WITH GRAY | DEEMPHASIZE WITH A LESS SATURATED HIGHLIGHT COLOR

But this comes with a disadvantage: All your not-highlighted categories will look the same. Your readers won't be able to distinguish between them if they're not directly labeled.

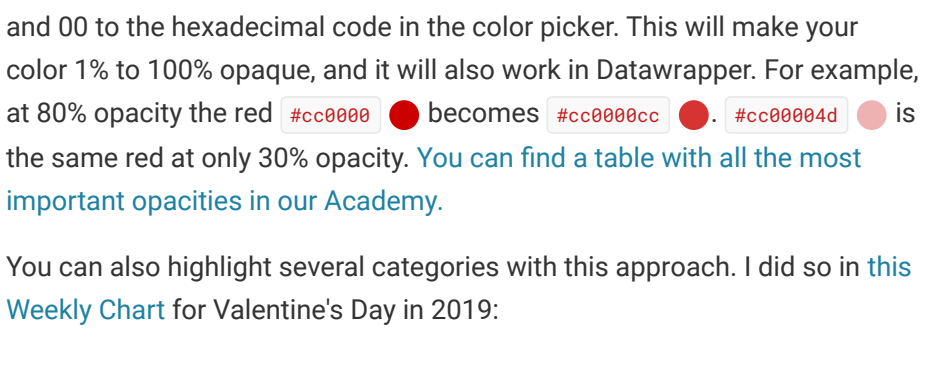
So when you have just a few categories (and not as many as Sarah in her scatter plot or the other The Economist designer in their line chart), consider using a distinctive hue for each category but making the less important ones less saturated:



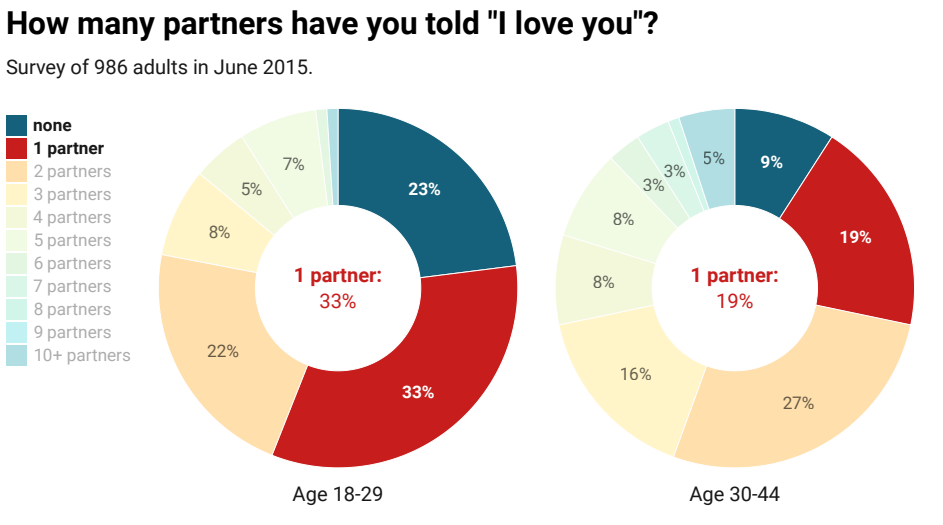
EVERYTHING IS EMPHASIZED | DEEMPHASIZE WITH THE SAME COLORS THAT ARE LESS SATURATED

Most **hover effects** in data vis do exactly that by reducing the opacity of everything that's not selected. "I get it, you want to select this category here," the visualization tells you as you hover.

You can use the same effect in your static visualizations. FiveThirtyEight and the Financial Times do so here:



FiveThirtyEight, 2021. The categories in red ● and blue ● are most important for the chart and hence more saturated than the light orange ●, light blue ●, and light green ●.



Financial Times, 2022. The BA.4/5 category ● is more important than the three other categories ●●●. The darker red makes it stand out despite being smaller.

A word about color technicalities: One easy way to make your colors less saturated is to make them less opaque (that is, more transparent). This will also make them brighter, which is often a better look than just reducing the saturation. For example, this red ● becomes almost brownish ● when we turn the saturation down, but looks like a nice pastel red ● when it's 80% opaque.

You can change the opacity of a color by adding a hex number between FF and 00 to the hexadecimal code in the color picker. This will make your color 1% to 100% opaque, and it will also work in Datawrapper. For example, at 80% opacity the red `#cc0000` ● becomes `#cc0000cc` ●. `#cc00004d` ● is the same red at only 30% opacity. You can find a table with all the most important opacities in our Academy.

You can also highlight several categories with this approach. I did so in this Weekly Chart for Valentine's Day in 2019:

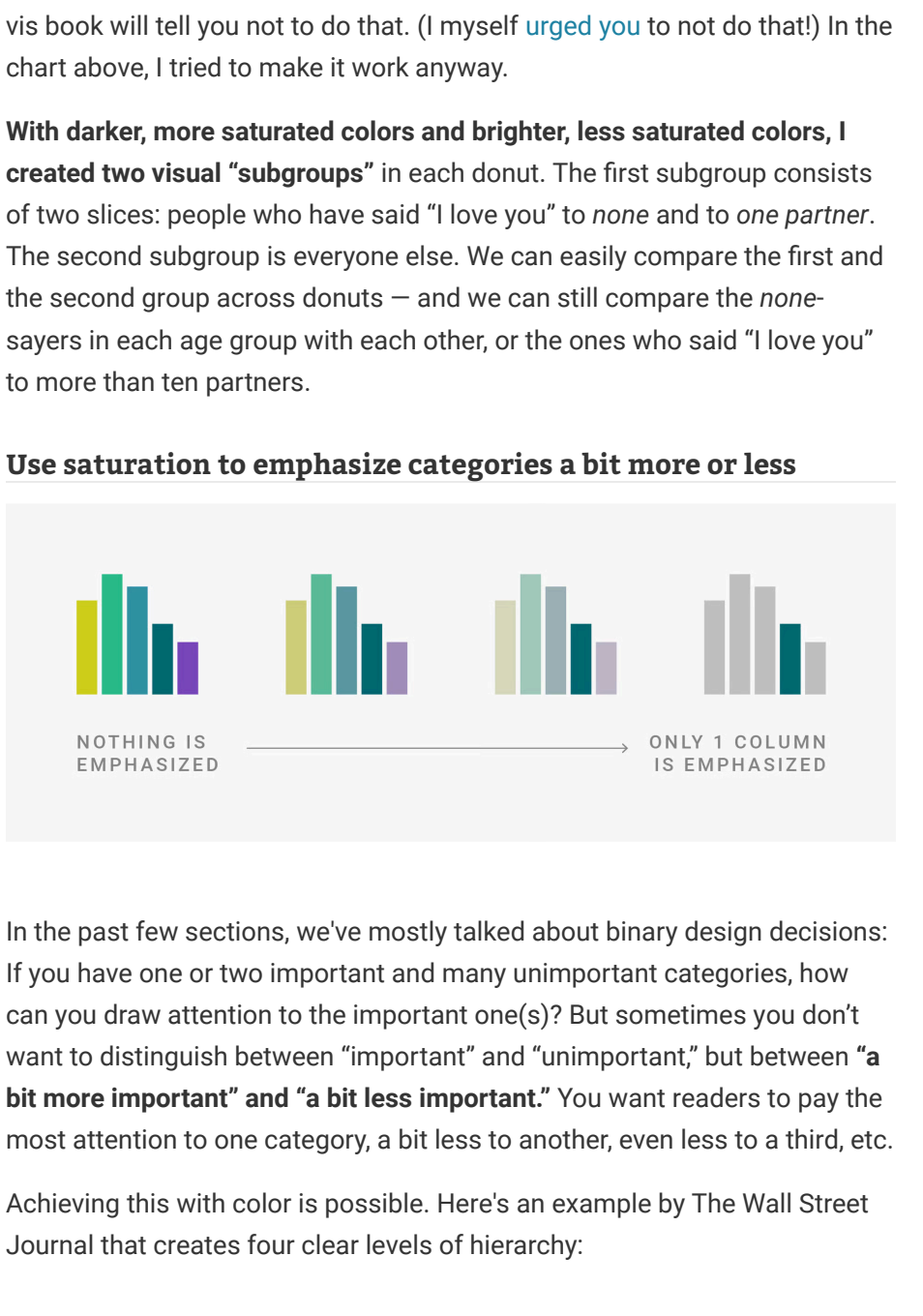
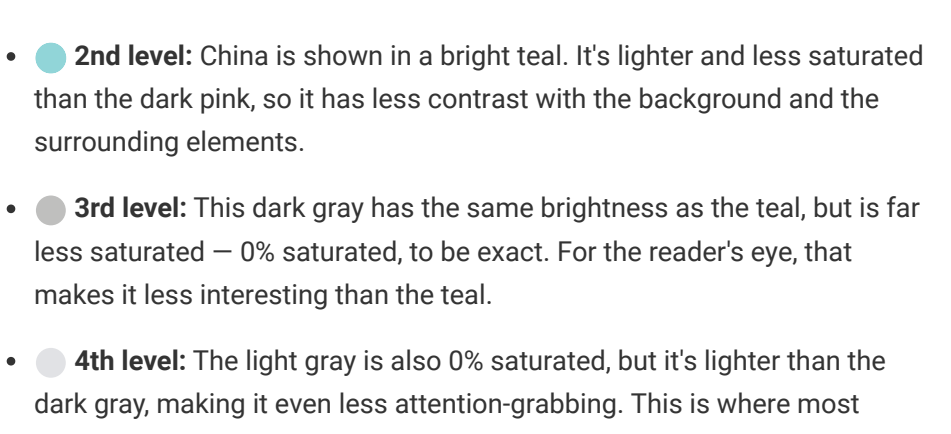


Chart: Lisa Charlotte Muth, Datawrapper • Source: YouGov

The chart above uses eleven different pie slices. Which is crazy! Every data vis book will tell you not to do that. (I myself urged you to not do that!) In the chart above, I tried to make it work anyway.

With darker, more saturated colors and brighter, less saturated colors, I created two visual "subgroups" in each donut. The first subgroup consists of two slices: people who have said "I love you" to *none* and to *one partner*. The second subgroup is everyone else. We can easily compare the first and the second group across donuts — and we can still compare the *none*-sayers in each age group with each other, or the ones who said "I love you" to more than ten partners.

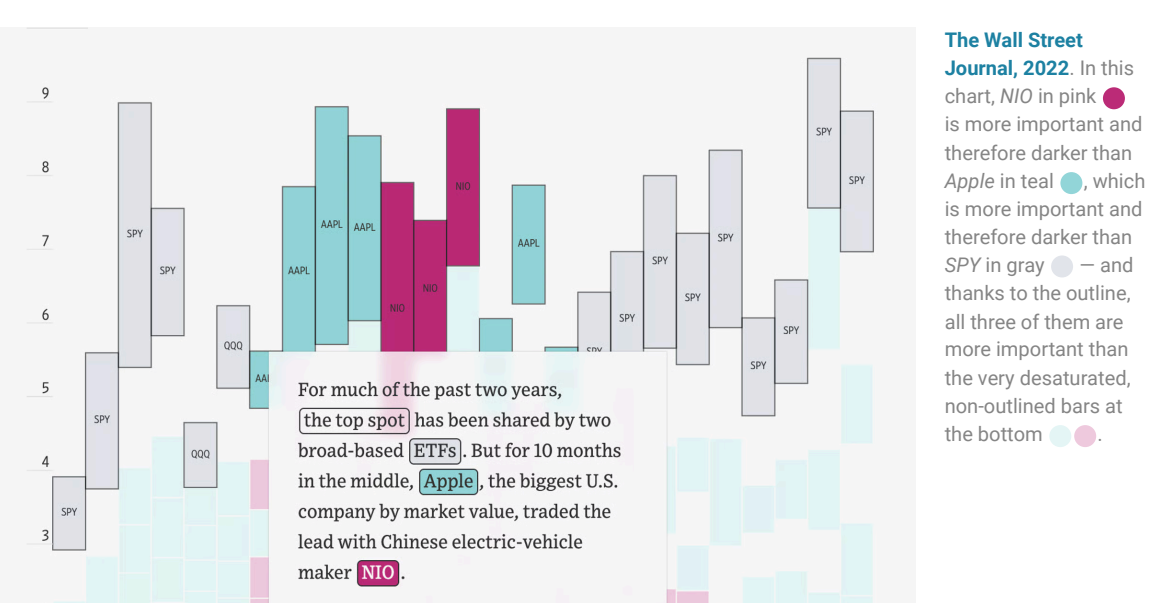
Use saturation to emphasize categories a bit more or less



NOTHING IS EMPHASIZED → ONLY 1 COLUMN IS EMPHASIZED

In the past few sections, we've mostly talked about binary design decisions: If you have one or two important and many unimportant categories, how can you draw attention to the important one(s)? But sometimes you don't want to distinguish between "important" and "unimportant," but between **"a bit more important" and "a bit less important."** You want readers to pay the most attention to one category, a bit less to another, even less to a third, etc.

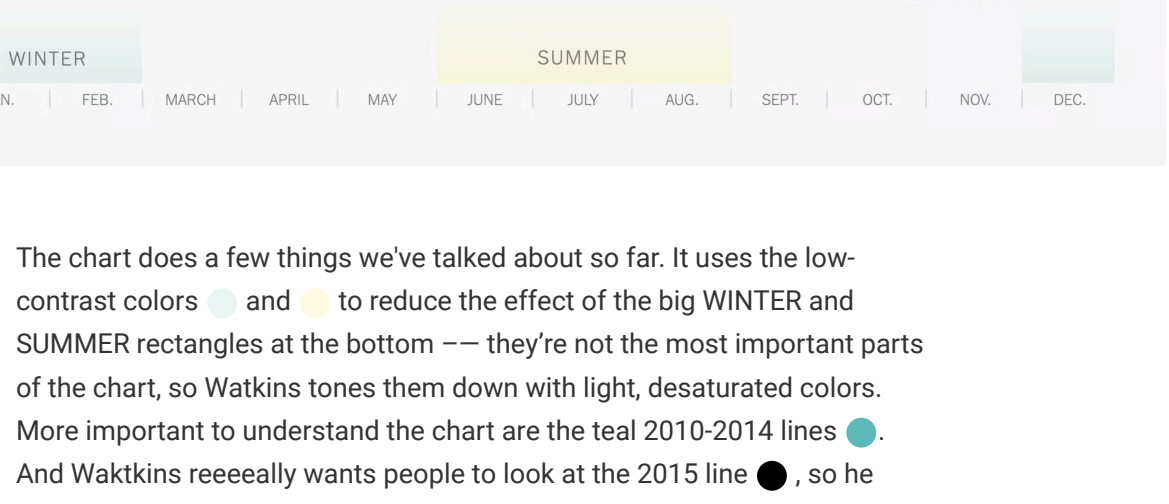
Achieving this with color is possible. Here's an example by The Wall Street Journal that creates four clear levels of hierarchy:



The Wall Street Journal, 2022.

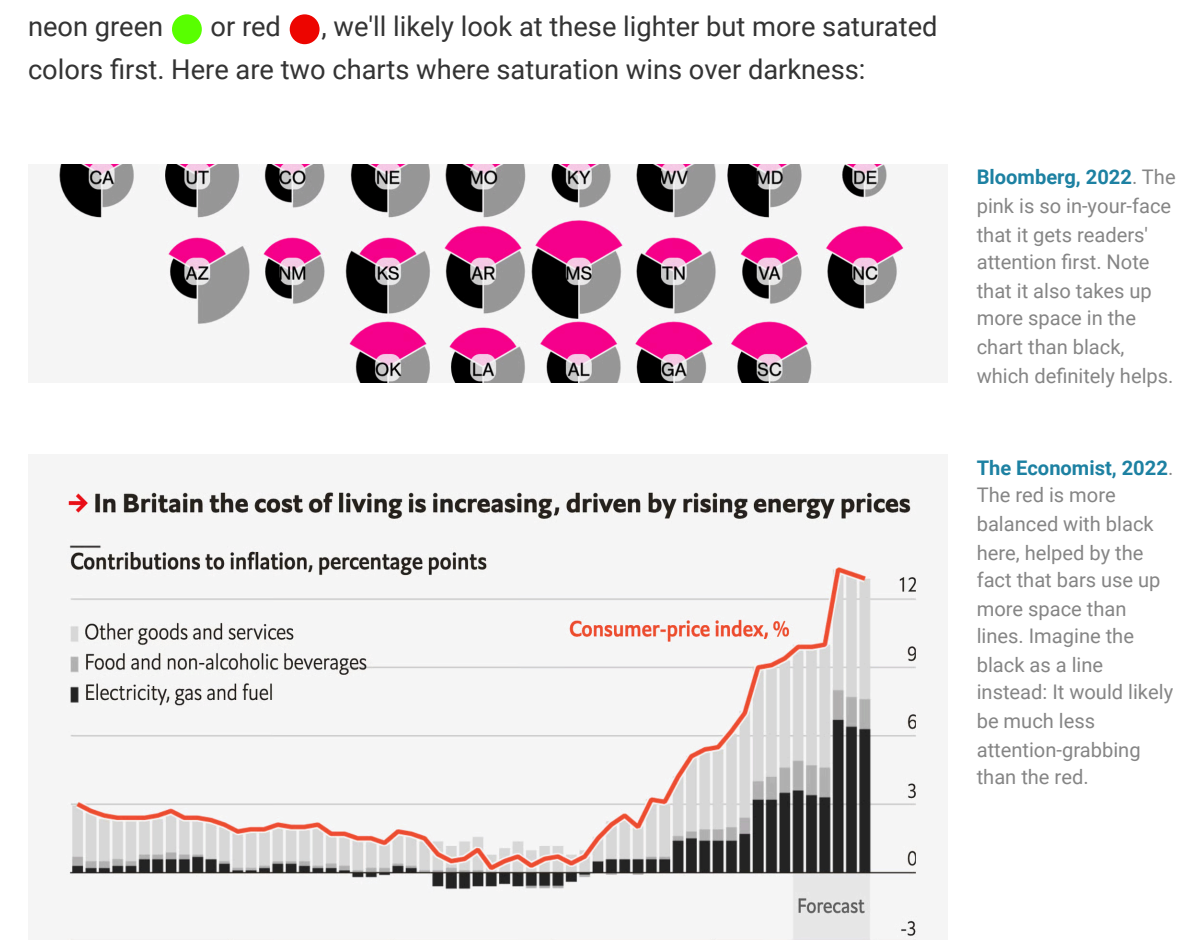
- ● **1st level:** The U.S. is shown in the most saturated and darkest bar color on this chart: dark pink. Most readers will immediately look at this category. In this case, the category doesn't cover many pixels, which makes it even more important that it's shown in such a high-contrast color.
- ● **2nd level:** China is shown in a bright teal. It's lighter and less saturated than the dark pink, so it has less contrast with the background and the surrounding elements.
- ● **3rd level:** This dark gray has the same brightness as the teal, but is far less saturated — 0% saturated, to be exact. For the reader's eye, that makes it less interesting than the teal.
- ● **4th level:** The light gray is also 0% saturated, but it's lighter than the dark gray, making it even less attention-grabbing. This is where most readers will look last.

As you can see, the **more saturated and darker (on a bright background) the colors, the more attention they'll get.** Here's another chart from the same article demonstrating that:



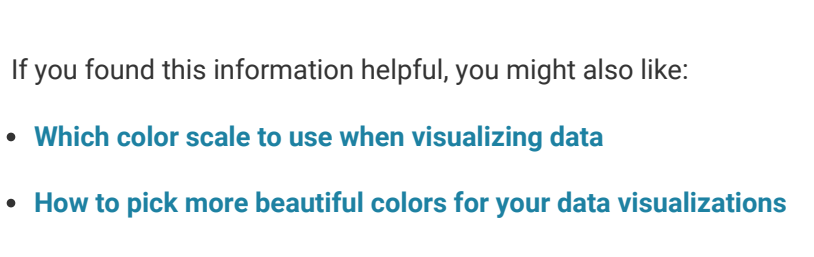
The Wall Street Journal, 2022. In this chart, *NIO* in pink ● is more important and therefore darker than *Apple* in teal ●, which is more important and therefore darker than *SPY* in gray ● — and thanks to the outline, all three of them are more important than the very desaturated, non-outlined bars at the bottom ●●.

Let's look at a more complex example (and at the same time one of my favorite data visualizations out there): This 2015 chart by Derek Watkins for The New York Times. The visualization uses six different colors:

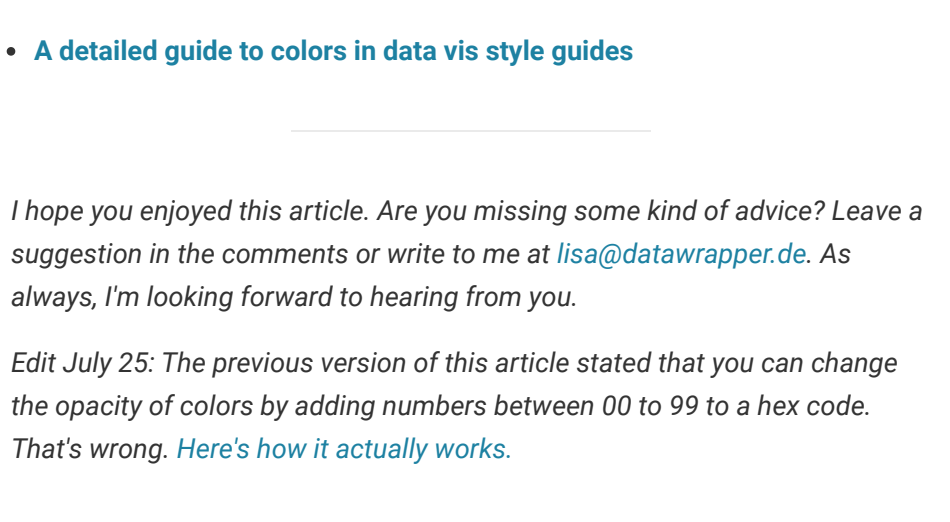


The chart does a few things we've talked about so far. It uses the low-contrast colors ● and ● to reduce the effect of the big WINTER and SUMMER rectangles at the bottom — they're not the most important parts of the chart, so Watkins tones them down with light, desaturated colors. More important to understand the chart are the teal 2010-2014 lines ●. And Watkins reeeeally wants people to look at the 2015 line ●, so he gives it the highest possible contrast (black on white) *and* makes it thicker.

Note that the darkest color — pure black — doesn't always win the race for most attention-getting. When seen together with a crazy-saturated color like neon green ● or red ●, we'll likely look at these lighter but more saturated colors first. Here are two charts where saturation wins over darkness:



Bloomberg, 2022. The pink is so in-your-face that it gets readers' attention first. Note that it also takes up more space in the chart than black, which definitely helps.



The Economist, 2022. The red is more balanced with black here, helped by the fact that bars use up more space than lines. Imagine the black as a fine instead: It would likely be much less attention-grabbing than the red.

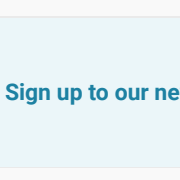
Choosing good colors is hard when you're just starting out — and it can take a long time even when you're more experienced. If you're not sure which of two colors will subtly catch your reader's eye more, just go back to creating a strong distinction between important and unimportant chart elements using "important" (dark, saturated) and "unimportant" (light, pastel, gray) colors.

If you found this information helpful, you might also like:

- **Which color scale to use when visualizing data**
- **How to pick more beautiful colors for your data visualizations**
- **10 ways to use fewer colors in your data visualizations**
- **A detailed guide to colors in data vis style guides**

I hope you enjoyed this article. Are you missing some kind of advice? Leave a suggestion in the comments or write to me at lisa@datawrapper.de. As always, I'm looking forward to hearing from you.

Edit July 25: The previous version of this article stated that you can change the opacity of colors by adding numbers between 00 to 99 to a hex code. That's wrong. Here's how it actually works.



Lisa Charlotte Muth (she/her, @lisacmuth, @lisacmuth@vis.social) is Datawrapper's head of communications. She writes about best practices in data visualization and thinks of new ways to excite you about charts and maps. Lisa lives in Berlin.

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COMMENTS

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Michiel Duvekot
commented on July 24, 2023 - 4:31 pm

I think there is an error in "you can change the opacity of a color by adding a number between 01 and 99". The range appears to be 0 to FF (0-255 in hexadecimal). If you want to make a colour like #cc0000 80% opaque, you would set it to #cc0000cc. (80% of 255 = 204)

> **Lisa Charlotte Rost**
> commented on July 25, 2023 - 8:05 am
>
> Hi Michiel, thanks for your comment and for pointing that out! I just changed it in the blog post and this Academy post.

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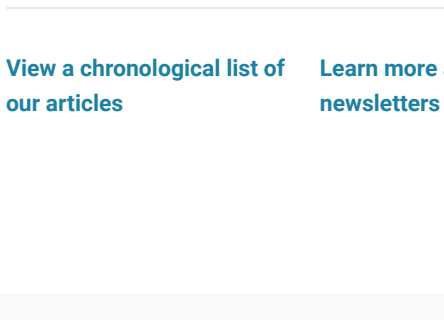

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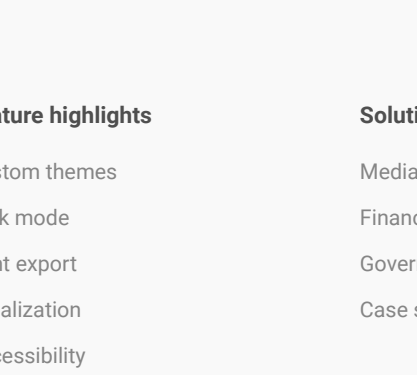

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